Streamlining Support: Automating Incident Classification and Routing with Flow Designer and Predictive Intelligence

Project Design Phase: Solution Architecture

Date:	October 3, 2026	**Maximum Marks:**	4 Marks
Team ID:	SWTID-2026-6328	**Project Name:**	Auto Ticket Classification

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

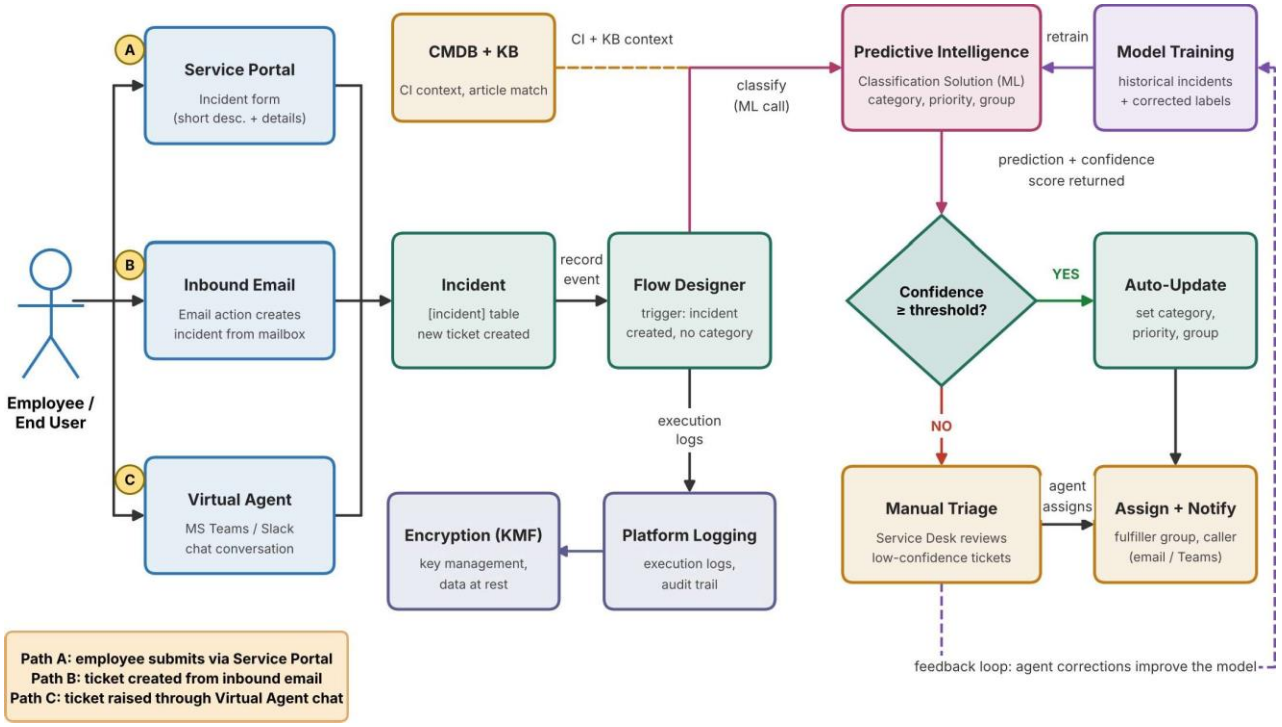


Figure 1: Architecture and data flow of the Auto Ticket Classification application

Architecture components:

The solution architecture for Auto Ticket Classification connects user touchpoints, an event-driven automation layer, and a machine-learning classification service into a zero-touch triage workflow. Every new incident is categorised, prioritised and assigned without manual effort when the model is confident, and falls back to a human queue when it is not.

User Interaction Layer (Frontend / Channels):

- **ServiceNow Service Portal:** Employees raise an incident using a simple form with a short description and details; category and priority are left for the system to fill in.
- **Inbound Email Actions:** Emails sent to the support mailbox are converted into incident records automatically.
- **Virtual Agent (MS Teams / Slack):** Conversational channel that captures the issue in chat and creates the incident behind the scenes.

Core Process & Automation Layer (ServiceNow Platform):

- **Incident Management (incident table):** Stores the ticket text, caller, configuration item and the fields to be classified.
- **ServiceNow Flow Designer (Workflow Engine):** Low-code orchestration layer triggered when an incident is created without a category. It calls the classification service, evaluates the result and branches the logic.
- **Confidence Threshold Decision:** Compares the model's confidence score with a configurable threshold to choose between automatic update and manual triage.

Intelligence Layer (Predictive Intelligence):

- **Classification Solution:** A trained ML model that predicts category, priority and assignment group from the short description and description text.
- **CMDB and Knowledge Context:** Optional configuration item and knowledge article data that enrich the prediction.
- **Model Training and Feedback:** The model is retrained on historical incidents and on corrections made by agents, so accuracy improves over time.

Fulfillment & Notification Layer (Backend / Operations):

- **Auto-Update:** When confidence is high, the flow writes the predicted category, priority and assignment group to the incident.
- **Manual Triage Queue:** Low-confidence tickets are placed in a service desk queue where an agent classifies them; the correction feeds back into training.
- **Assignment and Notifications:** Routes the ticket to the fulfiller group and notifies the caller and assignee by email or Teams.

Security & Audit Layer:

- **Platform Logging:** Execution logs and an audit trail of each flow run and each classification decision.
- **Encryption (Key Management Framework):** Protects data at rest and manages encryption keys.